Quantum mesh partitioning for heterogeneous devices: unequal work shares expose a structural limit of a varQITE ansatz

Team ZollerQu — Qollab × IonQ Hackathon, October 2026

Abstract. Quantum graph partitioning has so far been studied for *balanced* bipartitions, notably by IonQ with Ansys and Synopsys inside LS-DYNA. Modern clusters mix CPUs and GPUs, so a mesh must be split with an *unequal* target share α proportional to device speed. We formulate this heterogeneous split as an Ising problem and benchmark QAOA, its variants and IonQ's variational imaginary-time evolution (varQITE) against classical heuristics on nine reproducible finite-element meshes, grading every method against the exact optimum and counting every evaluation. The heterogeneous case changes the picture: varQITE with the published pair-rotation ansatz returns a near-optimal split on 78% of shots for equal devices but fails completely for α=0.8. We prove why: a qubit that only ever acts as the control of the ansatz's two-qubit gates keeps a 50/50 marginal for all angles, which caps the probability of a unique optimum at 1/2; equal shares hide this because their optimum is doubly degenerate. QAOA beats random sampling at both shares and matches local search at 12 qubits, but falls behind classical heuristics from 16 qubits on. No quantum advantage is claimed.

1 Introduction and contributions

Parallel PDE solvers partition their mesh among compute devices; the partition sets both the load balance and the communication volume. IonQ inserted quantum graph partitioning into Ansys/Synopsys LS-DYNA to reduce fill-in of sparse direct solvers, first with varQITE [1, 2] and later with Iterative-QAOA [3]; quantum annealing has been tried for HPC load balancing [4]. These studies use balanced halves and closed code. Our contributions:

1. **Heterogeneous partitioning**: a two-device split with target share $\alpha = s_B/(s_A + s_B)$ from device speeds, plus a device-choice layer for a synthetic multi-device pool.
2. **A structural limit of the varQITE ansatz under unequal shares** (Lemma 1), measured on all cases, and a one-layer change (A^+) that removes the cap.
3. **An open benchmark graded against the exact optimum**: nine meshes, five QAOA families, varQITE and five classical baselines, with total evaluations counted for every method and all tuning done on held-out meshes.
4. **Reproducibility**: meshes and region assignments shipped as data files; 30 injected bugs all caught by the checks; one batched IonQ job prepared.

2 Problem and model

Benchmark. Nine plates meshed by Delaunay triangulation (1 670–2 166 triangles, Fig. 1): a rectangle with a known answer, an L-plate with a heat source, two sources, a plate with a hole (stress-style mesh), and a five-step complexity ladder on one L-plate (uniform → three sources → heavy-material zone, 3× work → strong-coupling zone, 3× border cost → priority zone). Each mesh is grouped into n connected regions (one qubit each) by k-means with a connectivity repair; region work a_i and border cost c_{ij} are exact sums. Meshes and region assignments are stored as data files, because floating-point differences made 4 of 9 meshes differ between macOS/arm64 and Linux/x86-64 builds.

Ising model. With z_i=+1 (device A) or −1 (device B) and normalised $\tilde{a}_i$, $\tilde{c}_{ij}$,

$$E(z) = \sum_{\langle ij\rangle} \tilde{c}_{ij} \tfrac{1-z_i z_j}{2} + \lambda\Big(\sum_i \tilde{a}_i z_i - (1-2\alpha)\Big)^2 \; [+\mu P(z)],$$

the Ising form of graph partitioning [5] with a target share. For $\alpha \neq 0.5$ the balance term contains single-qubit fields $h_i \propto (1-2\alpha)\tilde{a}_i$, so $E(z) \neq E(-z)$ and the optimum is generically unique. The balance term couples every pair of regions ($n(n-1)/2$ two-qubit terms), which suits IonQ's all-to-all connectivity. Scenarios: equal devices (α=0.5, λ=10) and CPU+GPU with an assumed 4× faster GPU (α=0.8, λ=3.98); λ is shared per scenario and validated by exact enumeration.

Device choice (modelled). For a synthetic pool of 8 devices with assumed speed, memory and power, exhaustive search gives a fastest plan of 9.4 h / 27.3 kWh, an overnight plan (≤12 h) of 11.8 h / 23.7 kWh and a lowest-energy plan of 13.4 h / 22.0 kWh, against 37.5 h / 56.1 kWh for equal shares.

3 Methods

QAOA (noiseless statevector, cross-checked against Qiskit circuits to 10^{-15}): cost layer $e^{-i\gamma E}$, mixer $e^{+i\beta \sum X}$; linear-ramp LR-QAOA [6]; optimised angles with COBYLA, Nelder–Mead, Powell, L-BFGS-B (budget 200 evaluations) and shot-based SPSA [7]; CVaR-QAOA [8]; warm starts [9]; a symmetry-reduced variant at α=0.5. **varQITE** [2]: each step solves $G\dot{\theta} = d$ by least squares with $d_k = -(\langle P_k H\rangle - \langle P_k\rangle\langle H\rangle)$ and $G_{kj} = \partial\langle P_k\rangle/\partial\theta_j$ (parameter shift), starting from $|+\rangle^{\otimes n}$ with all angles zero. Ansätze: the paper's $e^{i\theta Z_r Y_q}$ on all pairs ordered by vertex weight (ZY); R_Y on the target followed by CX on all pairs or on graph edges (A); A with one R_Y per qubit in front (A^+, ours); an IonQ-native RZZ + RX ansatz (B). Training uses exact expectations. **Classical**: uniform and target-share random sampling, simulated annealing, local search with restarts, spectral bisection. **Metrics**: the probability per shot of a split with cost ≤ 1.05× optimum, and the probability that 1 000 shots or candidates contain one. **Total evaluations** count everything a method spends, in hardware-equivalent cost evaluations: (training circuits + 1) × 1 000 shots for quantum methods and the number of candidates for classical ones. **Honesty measures**: schedules, step sizes and annealing temperatures are tuned on separate calibration meshes; the p=0 control and the target-share sampler are always reported; a crude depolarising estimate $F=(1-0.005)^{N_{2q}}$ indicates noise sensitivity.

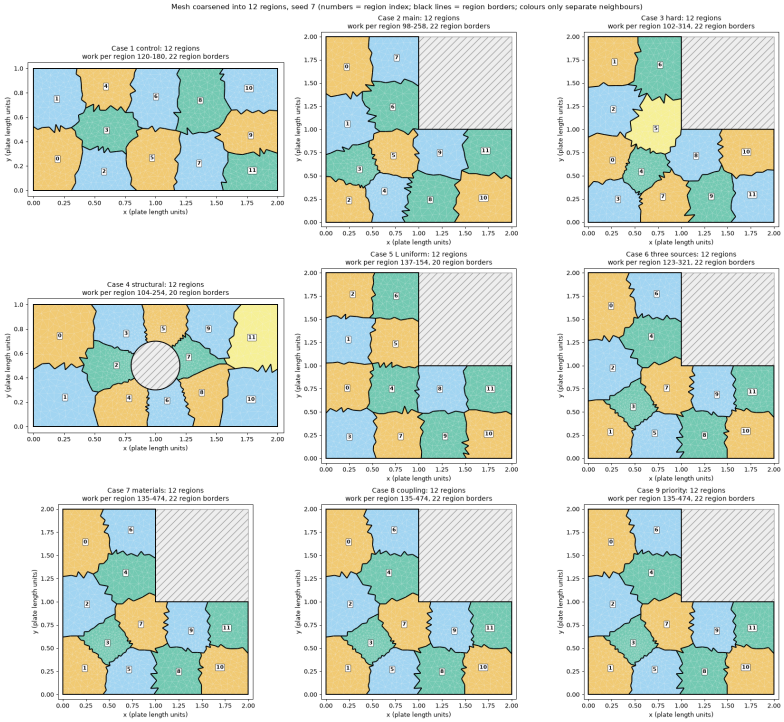


Figure 1: The nine benchmark meshes, each grouped into 12 regions (one qubit per region); numbers are region indices, black lines region borders.

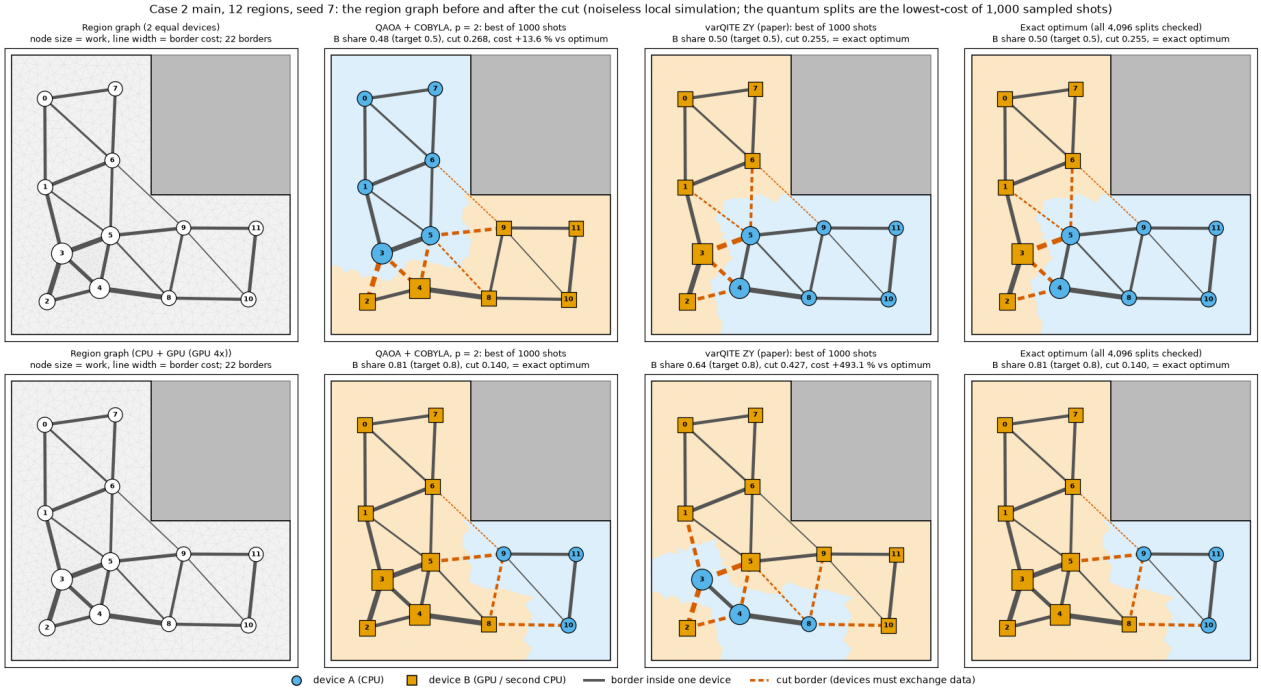


Figure 2: Case 2 (L-plate with a heat source), 12 regions. Columns: the region graph (node size = work, line width = border cost); the split returned by QAOA+COBYLA (p=2, best of 1 000 shots); the split returned by varQITE ZY; the exact optimum. Rows: equal devices (top), CPU+GPU with α=0.8 (bottom). varQITE finds the optimum for equal devices but not for the unequal share; QAOA's best shot is optimal for α=0.8.

4 The control-qubit limit of the varQITE ansatz

Lemma 1. *Let $U(\theta)$ be a product of gates that each commute with Z_r for a fixed qubit r (single-qubit gates on other qubits, CX with control r, $e^{i\theta Z_r Y_q}$, any gate diagonal on r). Then for every θ the measured marginal of r on $U(\theta)|+\rangle^{\otimes n}$ is exactly $1/2$, so every bitstring $x^\star$ has probability at most $1/2$.*

Proof. $P(x_r{=}1) = \langle\psi|\frac{1-Z_r}{2}|\psi\rangle$ with $\psi = U|+\rangle^{\otimes n}$; since $[U, Z_r] = 0$ this equals $\langle+|\frac{1-Z_r}{2}|+\rangle = \frac{1}{2}$, and $P(x^\star) \le P(x_r{=}x_r^\star) = \frac{1}{2}$. □

In the ZY ansatz ordered by vertex weight, the heaviest vertex is the Z side of every pair it touches; in ansatz A it is only ever a CX control. Both therefore contain such a qubit, which we verified numerically on all trained circuits. For α=0.5 without priority, $E(z){=}E(-z)$: the optimum comes as a pair $x^\star$, $\bar{x}^\star$, so a 50/50 qubit costs nothing, which is why the original MaxCut benchmarks did not expose it. For α=0.8 the optimum is unique and the cap binds. Adding one R_Y per qubit (A$^+$) breaks

Table 1: Results at n=12 qubits (nine cases, noiseless). Training circuits are means over cases and scenarios; total evaluations = (training + 1) × 1 000 for quantum methods, candidates for classical ones (spectral bisection: $2(n-1)$ threshold cuts). Per shot: probability of cost ≤ 1.05× optimum (%); hit@1000: probability that 1 000 shots/candidates contain such a split (%). †Starts from the classical spectral split.

				equal devices (α=0.5)		CPU+GPU (α=0.8)	
Method	2q gates	training circuits	total evaluations	per shot	hit@1000	per shot	hit@1000
Uniform random sampling	0	0	1 000	0.07	48	0.04	29
Random with target share	0	0	1 000	0.07	48	0.41	74
Simulated annealing	0	0	1 000	–	64	–	87
Local search (restarts)	0	0	1 000	–	84	–	100
Spectral bisection	0	0	22	–	67	–	100
LR-QAOA, $p = 2$	132	0	1 000	0.27	85	0.78	99
QAOA + COBYLA, $p = 2$	132	196	196 556	0.36	94	1.1	100
CVaR-QAOA + COBYLA, $p = 2$	132	31	31 778	0.31	90	0.87	99
Warm start (spectral)†, $p = 2$	132	200	201 000	7.8	94	21	100
varQITE ZY (paper, all pairs)	66	8 675	8 675 556	78	100	<0.01	<0.01
varQITE A (all pairs)	66	7 566	7 567 222	24	33	<0.01	0.01
varQITE A$^+$ (edges, our ext.)	22	5 003	5 004 222	<0.01	<0.01	<0.01	<0.01
varQITE B (IonQ-native, edges)	22	4 686	4 687 167	0.01	13	0.01	7.8

Table 2: hit@1000 (%) versus qubits n (cases 2, 9; seeds 7–9).

	equal devices				CPU+GPU			
n	12	16	20	24	12	16	20	24
Local search	86	45	18	10	87	96	77	52
Spectral bisection	50	33	50	67	83	100	100	100
Simulated annealing	65	32	16	12	82	48	39	26
Random, target share	39	5	0	0	65	17	8	9
LR-QAOA $p = 2$	76	20	1	–	99	50	12	–
QAOA+COBYLA $p = 2$	92	29	2	–	100	67	21	–

the commutation and removes the cap.

5 Results

12 qubits (Table 1, Fig. 2). All QAOA variants beat random sampling on every case: LR-QAOA raises the per-shot success 3.9× at α=0.5; at α=0.8 it is 22× uniform guessing but only 1.9× the target-share sampler (QAOA+COBYLA 2.6×). An ablation explains the gap: the single-region terms alone give 0.30% per shot, the pair terms alone 0.01%, both 0.78%. In hit@1000, QAOA+COBYLA (92%) edges out local search (86%) at α=0.5, but it spends 196 556 total evaluations against 1 000, and exhaustive search over all 4 096 splits is exact. Warm starts inherit most of their success from the spectral split; the convex-relaxation warm start is degenerate here (c_i=α for every region).

varQITE. At α=0.5 the ZY ansatz converges to near-optimal splits (78% per shot, 30–100% on every case) with 66 two-qubit gates, at a cost of 8 675 556 total evaluations. At α=0.8 every variant fails, as Lemma 1 predicts for A and ZY; A$^+$ removes the cap and reaches an approximation ratio of ≈ 0.98 but still misses the 5% window. At 16 qubits (graph-edge variants), A$^+$ reaches r=0.965 / 0.981 and no variant exceeds <0.01% per shot.

Scaling (Fig. 3, Table 2; 216 instances, 2 928 QAOA runs on an 8-core server). At fixed depth p=2 QAOA loses about an order of magnitude per shot for every four extra qubits. At 20 qubits QAOA+COBYLA reaches 2% / 21% (α=0.5 / 0.8), against 50% / 100% for spectral bisection and 18% / 77% for local search.

Table 3: The single IonQ batch: two-qubit gates and noiseless predictions (%) of P(optimum) and P(cost ≤ 1.05× opt.).

		equal		CPU+GPU	
Circuit	2q	opt	≤1.05	opt	≤1.05
p = 0 control (uniform)	0	0.05	0.05	0.02	0.05
LR-QAOA p = 1	66	0.13	0.13	0.31	0.60
LR-QAOA p = 2	132	0.14	0.14	0.46	0.89
QAOA + COBYLA p = 2	132	0.22	0.22	0.53	1.0
varQITE ZY (paper)	66	75	75	<0.01	<0.01
varQITE A$^+$ (edges)	22	<0.01	<0.01	<0.01	<0.01

Noise and outcome. The depolarising estimate lowers LR-QAOA (p=2, α=0.8) from 0.78% to 0.42% per shot and makes deeper circuits worse. The modelled runtime of every method's best split lies within 2.9% of the runtime-optimal split, so better samplers matter little for this objective at this size.

6 One batched IonQ job

Table 3 lists the 12 circuits prepared for a single IonQ Forte 1 noise-model job on Qollab (case 2, 12 qubits, 1 000 shots each): the p=0 control, LR-QAOA, QAOA+COBYLA and the final trained varQITE circuits. varQITE is trained on a noiseless local simulator; only its final circuits are sampled on IonQ. The Qollab program rebuilds the circuits from a data file, re-checks the exact optimum by brute force and grades every shot. The job has not been run yet.

7 Discussion and limitations

The heterogeneous share is not a cosmetic generalisation: it breaks the spin-flip symmetry that balanced benchmarks enjoy, and with it the success of a published ansatz. QAOA, whose mixer acts on every qubit, does not have this blind spot but is outpaced by classical heuristics beyond 12 qubits at fixed depth. All results are noiseless simulations or models; varQITE training used exact expectations; device speeds, runtimes and energies are assumptions. Next steps: run the IonQ batch; build single-qubit layers like A$^+$ into deeper varQITE ansätze; recursive splits for more than two devices; measured

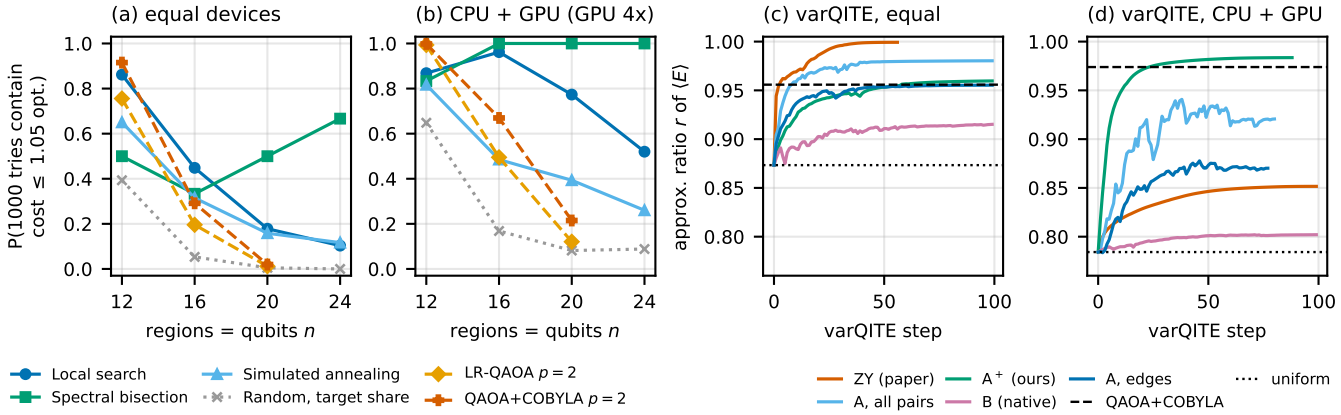


Figure 3: (a,b) Probability that 1 000 candidates (classical) or shots (QAOA, noiseless) contain a split within 5% of the optimum, versus problem size (cases 2 and 9, seeds 7–9; at 24 qubits the reference is the best classical result and QAOA is not simulated). (c,d) varQITE training curves at n=12 (mean over nine cases), with QAOA+COBYLA and uniform sampling for reference.

CPU/GPU runs on refined meshes.

This effort is supported by Qollab & IonQ. Code, data files and all results: `https://github.com/Oussema-t/zollerqu-qollab` (branch `mesh`). AI assistants (Claude Code, ChatGPT) were used for code and text; every number is produced by the published code.

References

[1] W. Aboumrad *et al.*, Accelerating large-scale linear algebra using variational quantum imaginary time evolution, arXiv:2503.13128 (2025).

[2] T. D. Morris, A. Kaushik, M. Roetteler, P. C. Lotshaw, Performant near-term quantum combinatorial optimization, arXiv:2404.16135 (2024).

[3] D. Zhu *et al.*, End-to-end performance of quantum-accelerated large-scale linear algebra workflows, arXiv:2603.15515 (2026).

[4] Load balancing for high performance computing using quantum annealing, arXiv:2408.04097 (2024).

[5] A. Lucas, Ising formulations of many NP problems, Front. Phys. **2**, 5 (2014).

[6] J. A. Montañez-Barrera, K. Michielsen, Toward a linear-ramp QAOA protocol, npj Quantum Inf. **11**, 131 (2025).

[7] J. C. Spall, Implementation of the simultaneous perturbation algorithm, IEEE Trans. Aerosp. Electron. Syst. **34**, 817 (1998).

[8] P. Kl. Barkoutsos *et al.*, Improving variational quantum optimization using CVaR, Quantum **4**, 256 (2020).

[9] D. J. Egger, J. Mareček, S. Woerner, Warm-starting quantum optimization, Quantum **5**, 479 (2021).